

College of Industrial Technology  
King Mongkut's University of Technology North Bangkok

Seat No.

Midterm Examination of Semester 2

Year: 2024

Subject: 031001112 Mechanics II

Section: 15-18

Date: 6 January 2025

Time: 10.00-12.00

Name: \_\_\_\_\_ ID: \_\_\_\_\_ Class: \_\_\_\_\_

Directions: The test is designed to measure your comprehension. The test is divided into 2 sections. There will be 5 pages (including this page) and they are worth 35 points.

- This exam paper contains no errors. If a suspected error is found, it is the student's discretion to correct it.
- Answer the questions on this test paper.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators cannot be used in this test.

Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

This exam paper is part of a set used to assess student abilities as follows:

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Part I | Part II |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------|
| CLO 1 | Explain, analyze and calculate various quantities related to projectile motion and explain the relationship between centripetal force radius of movement linear speed angular velocity and the mass of objects in circular motion in horizontal and horizontal planes, as well as calculating various quantities involved. and apply knowledge of circular motion Explaining the orbits of satellites.                                                                      | 1-3    | 1-4     |
| CLO 2 | Analyze and calculate the work of body force from the equation and area under the graph. Relationship between force and position Including explaining and calculating average power. Explain and calculate kinetic energy, potential energy, elastic potential energy, mechanical energy, and the relationship between work. Results force and kinetic energy law of conservation of energy Analyze quantities involved using the law of conservation of mechanical energy. | 4-5    | 5-7     |

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**Part I** Multiple choices. Choose the best answer for each question (1 point each)

1. Throw an object into the air and make the object move in a projectile motion. When the object falls to the same height as it was initially, which quantity is equal to zero?

- |                      |                          |
|----------------------|--------------------------|
| a. Acceleration      | b. Horizontal velocity   |
| c. Vertical velocity | d. Vertical displacement |

2. Mr. Off kicks a football with an initial velocity of 40 m/s at an angle of  $30^\circ$  with the horizontal. What is the vertical velocity of ball?

- |           |                      |
|-----------|----------------------|
| a. 0 m/s  | b. 20 m/s            |
| c. 40 m/s | d. Insufficient data |

3. Two stars with equal masses move in circles with different radius but the same period of motion. Which statement is correct?

- |                            |                                    |            |            |
|----------------------------|------------------------------------|------------|------------|
| A. Equal linear velocities | B. Equal angular velocities        |            |            |
| C. Equal frequencies       | D. Equal centripetal accelerations |            |            |
| a. A and C                 | b. B and D                         | c. C and D | d. B and C |

4. A child holds a bag with a force of 100 N and walks 10 m horizontally. Find the work that is done?

- |        |         |          |           |
|--------|---------|----------|-----------|
| a. 0 J | b. 10 J | c. 100 J | d. 1000 J |
|--------|---------|----------|-----------|

5. Which of the following has a negative work value?

- Work done by a force that makes an angle of 0 degrees with displacement.
- Work done by a force that makes an angle of 30 degrees with displacement.
- Work done by a force that makes an angle of 90 degrees with displacement.
- Work done by a force that makes an angle of 180 degrees with displacement.

**Part II Writing**

1. A ball is thrown horizontally from a height of 5 m. If the ball hits the ground 4 m from the bottom edge horizontally, find the initial velocity of the ball? (4 points)

2. A soccer player kicks a ball with an initial velocity of 50 m/s at an angle of  $53^\circ$  with the horizontal. Find how far from the starting point the ball lands on the ground? (5 points)

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3. Take an object of 1 kg and tie it to a string 1 m long and swing it in a horizontal circle with a constant speed of 10 m/s. Find

- Angular velocity, frequency, and period of motion? (answer in from of  $\pi$ ) (3 points)
- Centripetal force? (2 points)



4. A string is attached to a 0.5 kg ball that swings in a vertical circle of radius 1 m with a linear velocity of 3 m/s. Find the tension in the string at the lowest position of the object's motion? (4 points)

5. A man wants to move a 10 kg box 20 m horizontally in 10 s. He must exert a force of 100 N at an angle of  $60^\circ$  with the horizontal. Find the work and power done in moving the box? (4 points)

6. A 5 kg mass is attached to a spring suspended from the ceiling. When initially dropped from rest with the spring unstretched, the mass has a velocity of 2 m/s when dropped 0.2 m. Find the spring constant? (4 points)

7. The force on an object, acting along the x-axis, varies as shown in figure. Determine the work done by this force to move the object from  $x = 0$  to  $x = 30$  m? (4 points)

